AP Calc Lesson 1.1 Homework

Name ______________________

1. A cyclist is training for a bike race. On one Saturday, he records the following:
   - Between 8:00 AM and 9:30 AM, he rode 34 miles.
   - At 9:00 AM, the bike's speedometer read 27 miles per hour.

   a. What is the cyclist's average speed over the 1.5-hour interval?

   b. How does this compare to his instantaneous speed at 9:00 AM?

   c. Explain how a driver's speedometer relates to the concept of instantaneous rate of change.

2. A candle is burning in a quiet room. The height of the candle, $H(t)$, in centimeters, is measured at various times t , where t is the number of minutes since it was lit. The table below shows the recorded height at selected times.

t(minutes)	0	3	7	11
$H(t)$(cm)	12.0	11.6	10.9	10.2

   a. Find the average rate of change in the height of the candle between $t = 0$ and $t = 11$. Include units in your answer.

   b. Estimate the instantaneous rate of change in the candle's height at $t = 5$ minutes. Show your reasoning and include units in your answer.

3. The temperature in Queens, NY on a particular summer day is given by the function Q, where $Q(t)$ gives the temperature in degrees Fahrenheit, t hours after midnight. The temperature at midnight $(t=0)$ is 68°. The temperature at noon $(t=12)$ is 83°. Jenny, Kyle, and Dwaina are asked to estimate the rate at which the temperature is changing at 6 AM. Their responses are below.

Jenny: The rate is 0°/hour, since there can be no change in temperature at a single instant in time.

Kyle: The rate is undefined, since at exactly 6 AM, no time has passed, and the denominator of the rate of change expression would be 0.

Dwaina: The rate is defined and will be close to but not exactly equal to the average rate of change of the temperature between 6:01 and 5:59.

Who is correct? Explain.

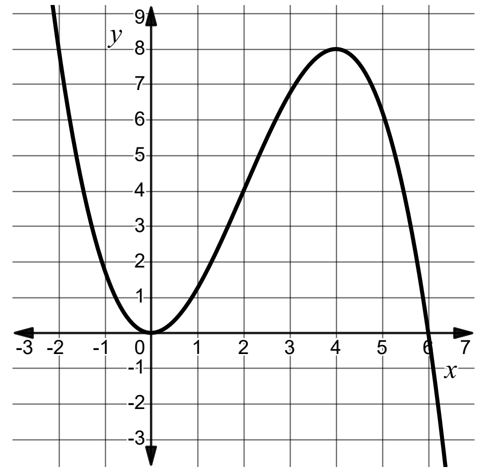
4. The number of customers at a coffee shop t hours after it opens can be modeled by $C(t) = -2(t-5)^2 + 50$ where $0 \leq t \leq 10$.

a. Find the average rate of change in the number of customer between $t=1$ and $t=8$.

b. Use the function to estimate the instantaneous rate of change at $t=3$ by calculating the average rate of change on the intervals $[3, 3.5], [3, 3.1], [3, 3.01]$ and $[3, 3.001]$.

c. What do you notice about the values of the average rate of change as the interval around 3 gets smaller?

5. The graph of $y = f(x)$ is shown.



a. Is the instantaneous rate of change of f greater at $x = 2$ or $x = 3$? How do you know?

b. Identify one x -value where the instantaneous rate of change of f is 0 .

c. Identify one interval where the average rate of change of f is 0 .

6. While downloading a large software update, Maya monitors her internet connection using a speed-tracking app. The app provides the total amount of data downloaded over time, as well as a real-time display of her current download speed. The total amount of data downloaded, in megabits (Mb), at different times is shown in the table below:

Time (seconds)	0	20	40	60	80	100
Data downloaded (megabits)	0	150	360	580	700	790

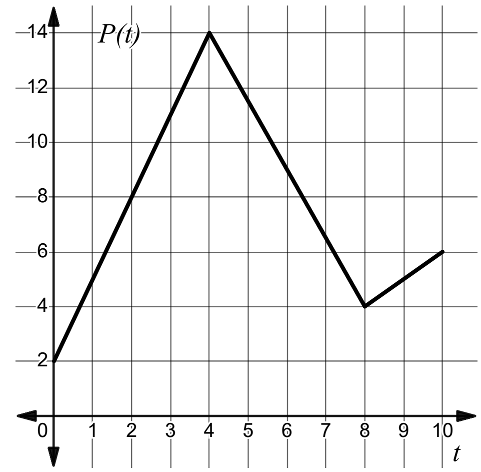
At the 60-second mark, the app displays a current download speed of 14 Mb/second, (commonly called Mbps).

a. Over which 20-second time interval is the download speed the fastest? How do you know?

b. Is the current download speed at the 60-second mark faster or slower than the average download speed on the interval $[20, 80]$? Show work to support your answer.

7. A robot vacuum moves back and forth along a long straight hallway. The robot's position, measured in meters from the charging station, is given by $P(t)$ where t is the time in minutes. The graph of P is shown for $0 \le t \le 10$.

a. Determine the instantaneous rate of change in the robot's position at $t = 5$.



b. What does your answer in part (a) indicate about the robot's movement?

8. Consider the graph of $f(x) = (x-2)^2 - 5$. Order the following from least to greatest.

I. The average rate of change of f on the interval $[-1, 5]$

II. The instantaneous rate of change of f at $x = 10$.

III. The instantaneous rate of change of f at $x = 8$.

IV. The average rate of change of f on $[2, 8]$.

V. The instantaneous rate of change of f at $x = 0$.

9. The number of cans of soup produced at a factory can be modeled by a function C, where $C(t)$ gives the number of cans produced, t hours after the factory begins their production lines at 5 AM. Which of the following is the best estimate of the instantaneous rate of change in the factory's soup can production at 7 AM?

A) $\dfrac{C(2)-C(2)}{2-2}$

B) $\dfrac{C(7)}{7}$

C) $\dfrac{C(2.1)-C(1.9)}{0.2}$

D) $\dfrac{C(3)-C(1)}{2}$

10. Let g be the functions given by $g(x)=\sqrt{x}-3x$. Find the average rate of change of g on the closed interval $[4,9]$.

A) $-\dfrac{34}{5}$

B) $-\dfrac{14}{5}$

C) $-\dfrac{8}{3}$

D) 2

AP Calc Lesson 1.2 Homework

Name ______________________

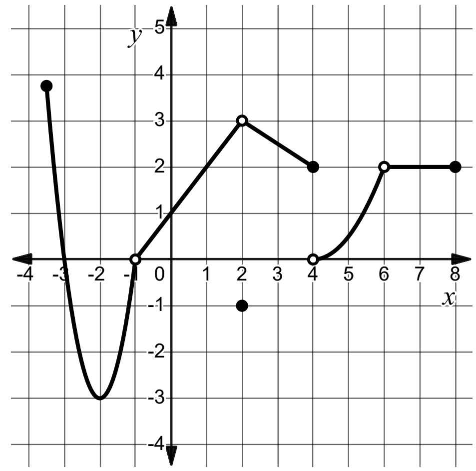
1. Use limit notation to describe each of the following:

a. As x approaches 3 , values of $f(x)$ approach -5 .

b. As x appraoches -2 from the right, values of $g(x)$ approach 10 .

c. The limit of $h(x)$ as x approaches 8 from the left is 0 .

2. Use the graph of $y = f(x)$ to evaluate each limit or state the limit does not exist.



a. $\lim\limits_{x \to -2} f(x)$

b. $\lim\limits_{x \to -1^-} f(x)$

c. $\lim\limits_{x \to -1^+} f(x)$

d. $\lim\limits_{x \to -1} f(x)$

e. $\lim\limits_{x \to 4^-} f(x)$

f. $\lim\limits_{x \to 4^+} f(x)$

g. $\lim\limits_{x \to 4} f(x)$

h. $\lim\limits_{x \to 6} f(x)$

i. $\lim\limits_{x \to 2} f(x)$

3. Selected values of a function f are given in the table. Estimate $\lim\limits_{x \to 5} f(x)$.

x	4	4.5	4.9	4.99	4.999	5.001	5.01	5.1
$f(x)$	5.104	5.820	6.715	6.970	6.997	7.003	7.030	7.316

4. Let $g(x) = \begin{cases} 2\sin\left(\frac{\pi}{2}x\right) - 3 & x < 3 \\ x^2 - 4x + 2 & x > 3 \end{cases}$. Evaluate $\lim_{x \to 3} g(x)$ or explain why the limit does not exist.

5. Write the equation of a quadratic function f where $\lim_{x \to -5} f(x) = 0$ and $\lim_{x \to 0} f(x) = -15$.

6. For a function f it is known that $\lim_{x \to 7} f(x) = -1$. Determine if each statement is guaranteed to be true, might be true, or guaranteed to be false.

a. $\lim_{x \to 7^-} f(x) = -1$

b. $f(7) = -1$

c. $\lim_{x \to 7^+} f(x)$ does not exist

d. $\lim_{x \to 7^-} f(x) = \lim_{x \to 7^+} f(x)$

e. $\lim_{x \to 7^+} f(x) = f(7)$

7. Evaluate each of the following limits or explain why they do not exist.

a. $\lim\limits_{x \to -1} (3x^2 - 4x + 5)$

b. $\lim\limits_{x \to 4} \dfrac{1}{x-4}$

c. $\lim\limits_{x \to 3} \sqrt{-2x+10}$

8. Let $f(x) = \begin{cases} \ln(x-1) + 3x & x < 2 \\ -2x + b & x > 2 \end{cases}$ for some constant b.

a. Find the value of b so that $\lim\limits_{x \to 2} f(x)$ exists or explain why no such value exists.

b. Find the value of b so that $\lim\limits_{x \to 2} f(x)$ does not exist or explain why no such value exists.

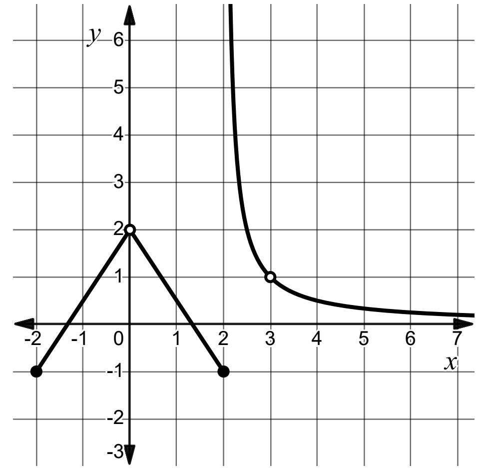
9. The graph of the function $y = f(x)$ is shown. Which of the following limits does not exist?

A) $\lim_{x \to 0^+} f(x)$

B) $\lim_{x \to 2^-} f(x)$

C) $\lim_{x \to 3} f(x)$

D) $\lim_{x \to 2} f(x)$



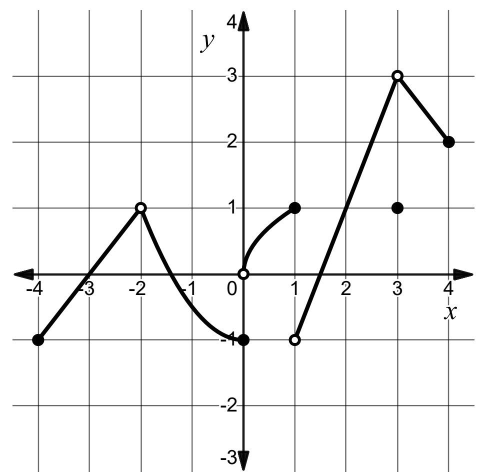
10. The graph of the function $y = h(x)$ is shown. If $\lim_{x \to a} h(x) = 1$, which of the following could be the value of a?

A) 1 and 3

B) -2 and 2

C) 2 only

D) $-2, 1, 2$ and 3



AP Calc Lesson 1.3 Homework

Name ______________________

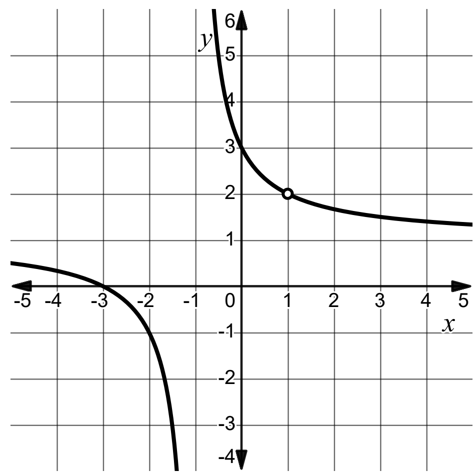
1. Evaluate $\lim\limits_{m \to 4} \dfrac{(-2m+7)(m-4)}{m-4}$.

2. Evaluate $\lim\limits_{x \to -6} \dfrac{x^2+10x+24}{x^2+5x-6}$.

3. The graph of $y = g(x)$ is given.

   a. Use a limit statement to describe the behavior of g at $x = 1$.

   b. Use two limit statements to describe the behavior of g at $x = -1$.



4. Let $h(x) = \dfrac{x^2-11x-26}{x^2-7x+10}$. For which value(s) of a does $\lim\limits_{x \to a} h(x)$ not exist?

5. Let $f(x) = \dfrac{(x-a)}{(x-b)(x-c)}$ for some constants $a, b,$ and c . It is known that $\lim\limits_{x \to -5} f(x)$ does not exist, $\lim\limits_{x \to 3} f(x)$ exists, $f(3)$ is undefined, and $b < c$. Find the values of $a, b,$ and c .

6. Let g be the function given by $g(x) = \dfrac{x-3}{x^4-81}$. How many vertical asymptotes does the graph of g have? Show your reasoning.

7. Let $f(x) = \dfrac{x^2-a}{x^2+bx+12}$ for some constants a and b . If the graph of f has a hole at $x=-4$, find the value of a and b .

8. Write the equation of a rational function f that meets the following requirements:
   - Domain: $(-\infty,-4)\cup(-4,0)\cup(0,\infty)$
   - $\lim_{x\to 8} f(x) = 0$
   - $\lim_{x\to 0} f(x) = -32$

9. The graph of f has a vertical asymptote given by $x=6$. Which of the following statements is guaranteed to be false?

   A) $\lim_{x\to 6} f(x) = \infty$

   B) $\lim_{x\to 6^-} f(x) = -\infty$

   C) $\lim_{x\to 6} f(x) = 0$

   D) $\lim_{x\to \infty} f(x) = 6$

10. Which of the following statements is true about $h(x) = \dfrac{x^2-12x+35}{x^2+2x-63}$?

    A) $\lim_{x\to -9} h(x)$ and $\lim_{x\to 7} h(x)$ do not exist.

    B) $\lim_{x\to -9} h(x)$ does not exist but $\lim_{x\to 7} h(x)$ exists.

    C) $\lim_{x\to 7} h(x)$ does not exist but $\lim_{x\to -9} h(x)$ exists.

    D) $\lim_{x\to 5} h(x)$ exists but $\lim_{x\to 7} h(x)$ does not exist.

AP Calc Lesson 1.4 Homework

Name ______________________

1. Let $f(x) = -5x^3 + 8x + 1$ and $g(x) = e^{2x} - 20x$.

   a. Evaluate $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)}$.

   b. Evaluate $\lim\limits_{x\to\infty} \dfrac{g(x)}{f(x)}$.

2. To reduce noise complaints, a city installs sound-dampening panels along a major highway. The average decibel level of traffic noise measured x meters away from the highway is given by $D(x)$, where $x \geq 0$. It is known that $\lim\limits_{x\to\infty} D(x) = 45$. Interpret this statement in the context of this problem.

3. Evaluate $\lim\limits_{x\to\infty} \dfrac{\ln(3x^2 - 5)}{2x + 8}$.

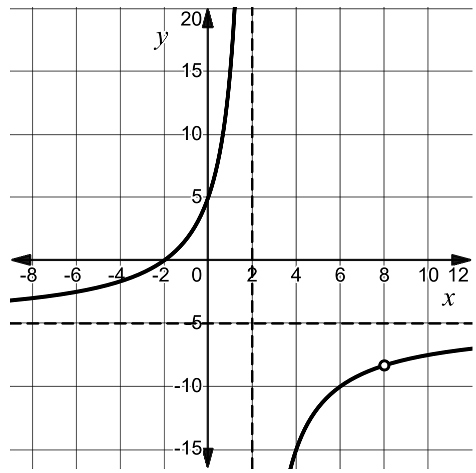
4. Evaluate $\lim\limits_{x\to\infty} \dfrac{8x^2 - 8x + 2}{\sqrt{25x^4 + 3x^2 + 2}}$.

5. Let $h(x) = \dfrac{f(x)}{7x^2 - 3x^4 + 5}$ for some function f . If $\lim\limits_{x\to\infty} h(x) = 2$, write a possible equation for $f(x)$.

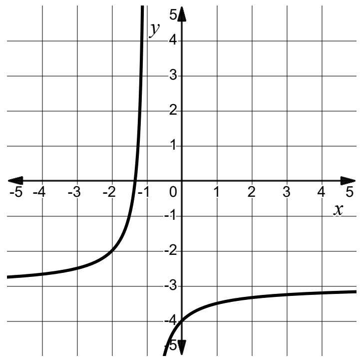
6. Write the equation of a rational function g such that $\lim\limits_{x\to 4} g(x) = 0,$ $\lim\limits_{x\to -3} g(x)$ does not exist and $\lim\limits_{x\to\infty} g(x) = 2$.

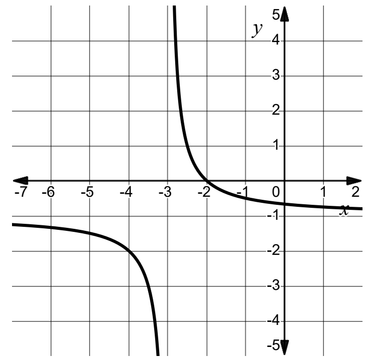
7. Let $f(x) = \dfrac{(2x-6)(8-3x)}{(3x-1)^2}$. Write the equation of the horizontal asymptote of f .

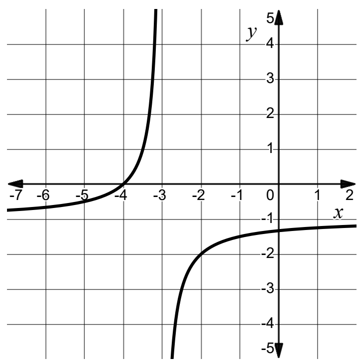
8. Let h be a function such that $h(x) = \dfrac{(ax-b)(x+2)}{x^2+cx+d}$ for constants $a, b, c,$ and d. The graph of $y = h(x)$ is shown. Find the values of $a, b, c,$ and d .

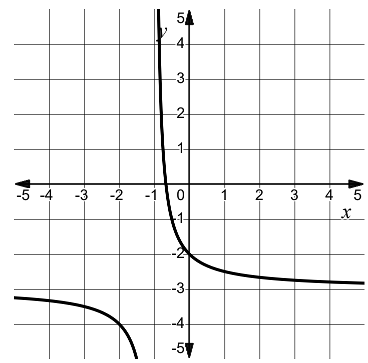


9. Let f be a rational function with $\lim\limits_{x\to\infty} f(x) = -3$, $\lim\limits_{x\to -1^+} f(x) = -\infty$, and $\lim\limits_{x\to -1^-} f(x) = \infty$. Which of the following could be the graph of f?

A) 

B) 

C) 

D) 

10. The function g is given by $g(x) = \cos\left(\dfrac{2x-1}{x^3}\right)$. Which of the following statements are true?

I. $\lim\limits_{x\to -\infty} g(x) = -1$

II. $\lim\limits_{x\to 0} g(x)$ does not exist

III. $\lim\limits_{x\to \infty} g(x) = 1$

A) I only

B) II only

C) II and III only

D) I, II, and III

AP Calc Lesson 1.5 Homework

Name ______________________

1. Consider the function given by $f(x) = \dfrac{|x+5|}{x+5}$.

a. Evaluate $\lim\limits_{x \to -5^-} f(x)$.

b. Evaluate $\lim\limits_{x \to -5^+} f(x)$.

c. Evaluate $\lim\limits_{x \to -5} f(x)$.

2.

x	1.9	1.99	1.999	1.9999	2	2.0001	2.001	2.01	2.1
$f(x)$	2.915	2.938	2.961	2.996	11	3.004	3.039	3.062	3.085

The table gives values of functions f at selected values of x . It is known that $\lim\limits_{x \to 2} f(x)$ exists.

a. Evaluate $\lim\limits_{x \to 2} f(x)$.

b. If $\lim\limits_{x \to 2} g(x) = 4$, find $\lim\limits_{x \to 2} (7g(x) - f(x))$.

3. If f is the function defined by $f(x) = \dfrac{x^2 - 16}{\sqrt{x} - 2}$, then $\lim\limits_{x \to 4} f(x)$ is

4. Let $f(x) = \dfrac{|x|}{x}$. Evaluate $\lim\limits_{x \to 2^+} [-5f(x-2)]$.

5. Given that $\lim_{x\to -1} f(x) = 6$ and $\lim_{x\to -1} g(x) = -2$, find the following limits.

a. $\lim_{x\to -1^+} \frac{f(x)}{4}$

b. $\lim_{x\to -1} \frac{f(x)}{g(x)}$

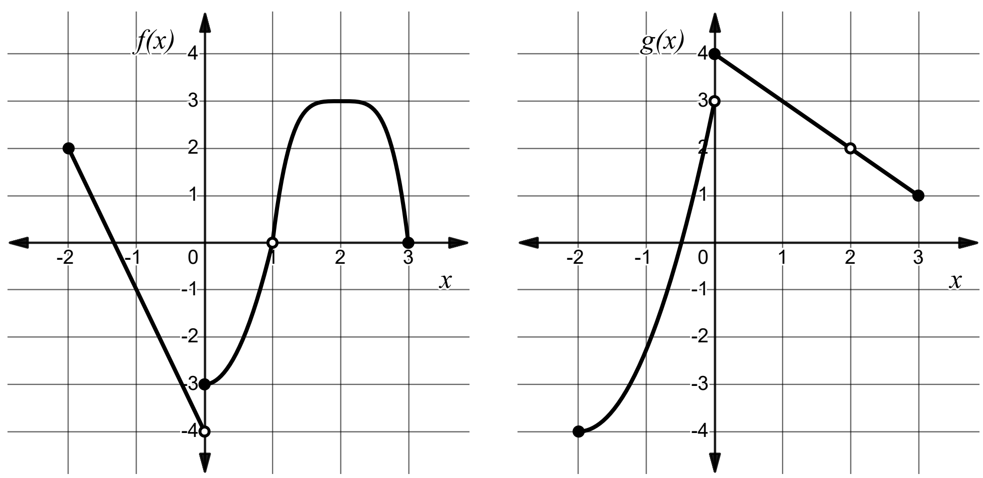
c. $\lim_{x\to -1} [f(x) \cdot g(x)]$

6. Let f be the function defined by $f(x) = \frac{\sin(kx)}{4x}$, where k is a nonzero constant. For what value of k , if any, does $\lim_{x\to 0} f(x) = 4$?

7. The graphs of functions f and g are shown in the figures. Evaluate each of the following.

a. $\lim_{x\to 0} [2g(x+1)]$

b. $\lim_{x\to 0} [f(x) - g(x)]$



c. $\lim_{x\to 2} [f(g(x))]$

8. What is $\lim_{x\to 0} \frac{\tan(4x)}{3x\sec(6x)}$?

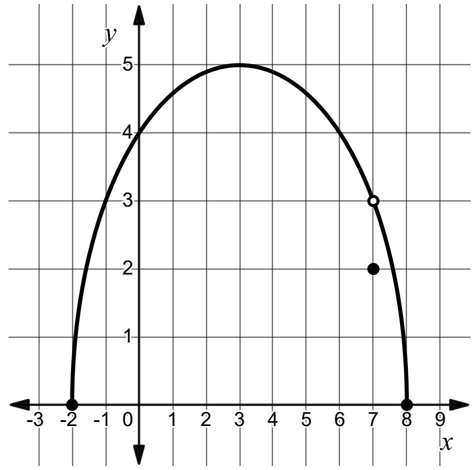
9. The figure shows the graph of $y = f(x)$. Evaluate $\lim_{x \to 7} [f(f(x))]$.

A) 2

B) 3

C) 5

D) The limit does not exist.



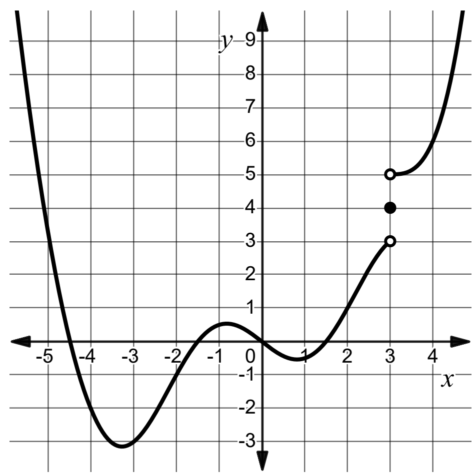
10. The figure shows the graph of $y = f(x)$. Find the value of $\lim_{x \to 0} f(3 + x^2)$.

A) 3

B) 4

C) 5

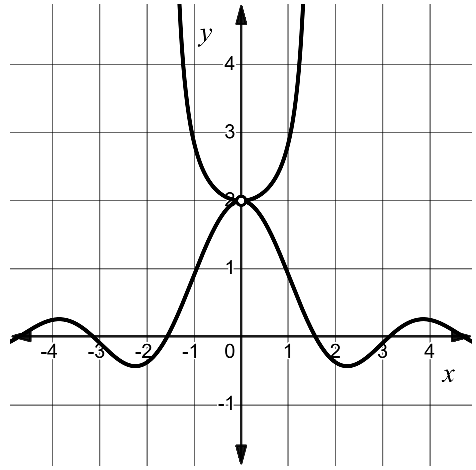
D) The limit does not exist.



AP Calc Lesson 1.6 Homework

Name ______________________

1. The graphs of functions g and h are shown in the figure. It is known that $g(x) \leq f(x) \leq h(x)$ for all x . Evaluate $\lim_{x \to 0} f(x)$.

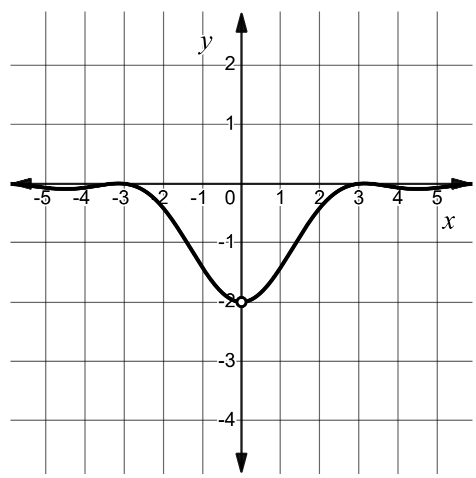


2. Let g and h be the functions defined by $g(x) = 4x - \frac{1}{3}x^2$ and $h(x) = x^2 - 2x + 7$. Function f satisfies $g(x) \leq f(x) \leq h(x)$ for all x . Determine $\lim_{x \to 3} f(x)$ or explain why this is not possible.

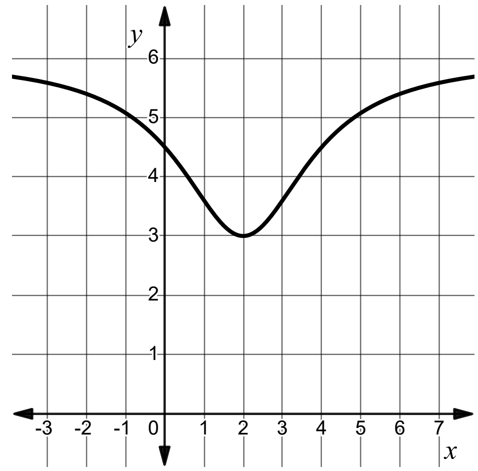
3. Let f be the function given by $f(x) = \frac{\cos(2x) - 1}{x^2}$. Function f satisfies $g(x) \leq f(x) \leq h(x)$.

a. Find a rule for g that provides an appropriate lower bounding function.

b. Find a rule for h that provides an appropriate upper bounding function.



4. Let f, g, and h be functions that satisfy $g(x) \leq f(x) \leq h(x)$ for all x . Function g is given by $g(x) = 3 - (x-2)^4$ and the graph of $y = h(x)$ is shown in the figure.
Evaluate $\lim_{x \to 2} f(x)$ or explain why the limit cannot be found.



5.

x	3.9	3.99	3.999	3.9999	4	4.0001	4.001	4.01	4.1
$h(x)$	6.198	6.159	6.034	6.002	Undefined	6.002	6.034	6.159	6.198

The table gives values of function h at selected values of x . Function h is continuous for all $x \neq 4$. Function g is given by $g(x) = \dfrac{12}{x^2 - 8x + 18}$. If $g(x) \leq f(x) \leq h(x)$ evaluate $\lim_{x \to 4} f(x)$ or explain why this is not possible.

6. Consider the function f given by $f(x) = \dfrac{2 - \cos(x)}{x+3}$.

a. Identify the range of $y = \cos(x)$.

b. Identify the range of $y = 2 - \cos(x)$.

c. Use your answer to part b to find functions g and h such that $g(x) \leq f(x) \leq h(x)$.

d. Evaluate $\lim_{x \to \infty} \dfrac{2 - \cos(x)}{x+3}$ using the Squeeze Theorem.

7. Evaluate the given limit using the Squeeze Theorem.

$$\lim_{x\to 0}\left[x^4\sin\left(\frac{6}{x}\right)-2\right]$$

8. Let f, g, and h be functions such that $f(x)\le g(x)\le h(x)$ for all x in the interval $-1\le x\le 8$. The Squeeze Theorem was used to determine that $\lim_{x\to 5} g(x)=12$. Which of the following statements is true about the functions f and h ?

A) $\lim_{x\to 5} f(x)=11$ and $\lim_{x\to 5} h(x)=13$

B) $\lim_{x\to 5} f(x)<12$ and $\lim_{x\to 5} h(x)>12$

C) $\lim_{x\to 5} f(x)=\lim_{x\to 5} h(x)=12$

D) There is not enough information to determine whether $\lim_{x\to 5} f(x)$ or $\lim_{x\to 5} h(x)$ exist.

9. The function f is defined by $f(x)=\dfrac{2\cos(x)-2}{x^2}$ for all $x\neq 0$. The inequalities given are true for $x\neq 0$ on the interval $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$. Which of the following inequalities can be used to show that $\lim_{x\to 0} f(x)=-1$ by the Squeeze Theorem?

I. $-x^2-1\le f(x)\le x^2-1$
II. $-\sec(x)\le f(x)\le -\cos(x)$
III. $-|x|-1\le f(x)\le |x|-1$

A) I and II only

B) I and III only

C) II and III only

D) I, II, and III

AP Calc Lesson 1.7 Homework

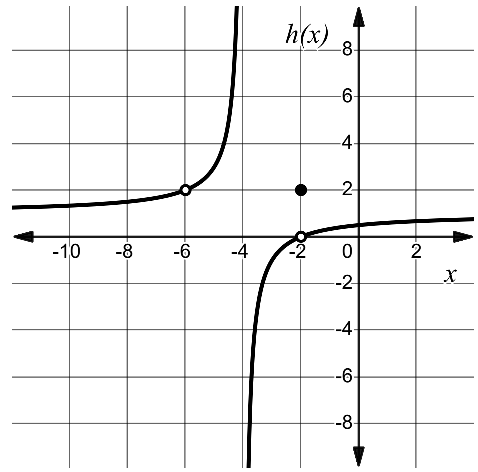
Name ______________________

1. The graph of function h is shown in the figure.

   a. At which value(s) of x does the graph of h have a removable discontinuity?

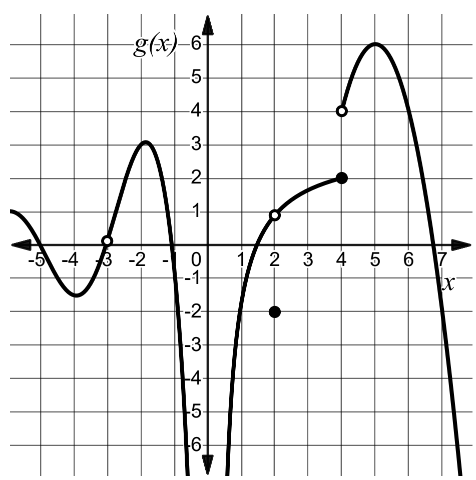
   b. At which value(s) of x does the graph of h have an infinite discontinuity?

   c. On which intervals is h continuous?



2. Let function f be given by $f(x) = \dfrac{x^2 - 8x + 12}{x^2 + 3x - 10}$. At which x -value does f have a removable discontinuity?

3. The graph of a function g is shown in the figure. At which x -values is g not continuous. State the type of discontinuity.



4. For a function g , it is known that $\lim\limits_{x \to 2^-} g(x) = 8$ and $\lim\limits_{x \to 2^+} g(x) = 8$. Does this guarantee that g is continuous at $x = 2$? Explain why or why not.

5. Let f be a function defined by $f(x) = \begin{cases} 4\cos(\pi x) & x < 2 \\ x^2 - \ln(x-1) & x \geq 2 \end{cases}$. Show that f is continuous at $x = 2$ using the definition of continuity.

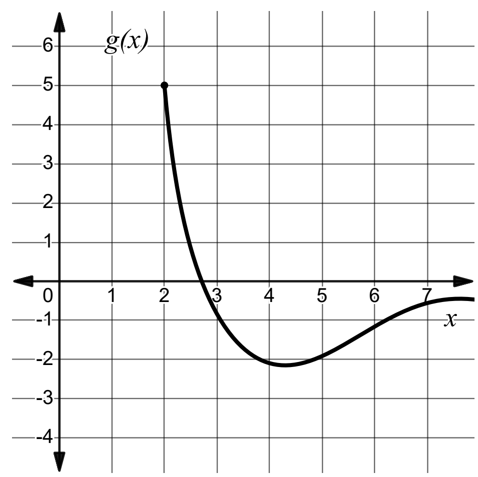
6. Consider the graph of $f(x) = \dfrac{|x+7|}{x+7}$.

a. Over which interval(s) is the graph of f continuous?

b. For which x-values is the graph of f discontinuous? Identify the type of discontinuity and justify your answer with limits.

7. The piecewise function f is defined below. The graph of $y = g(x)$ is given in the figure. Determine whether f is continuous at $x = 2$. Justify your answer.

$$f(x) = \begin{cases} 3x^2 - 5x + 1 & x < 2 \\ g(x) & x \geq 2 \end{cases}$$



8. Let f be the function given by $f(x) = \dfrac{(x+3)(ax-8)}{x^2+bx+c}$. If function f has a removable discontinuity at $x=2$ and an infinite discontinuity at $x=4$, find the values of $a, b,$ and c.

9. Let f be a continuous function that satisfies $f(4)=10$. Which of the following limits is guaranteed to be true?

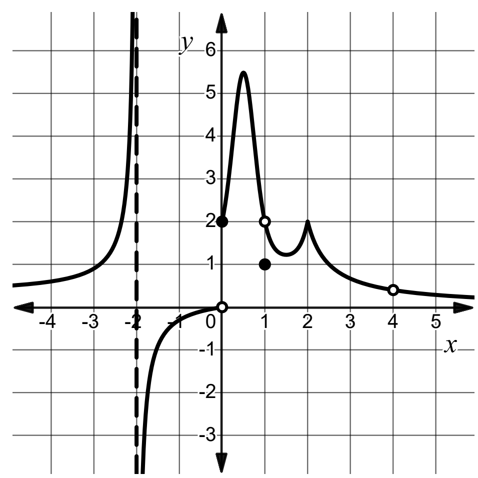
A) $\lim\limits_{x\to 2} f(2x) = 5$

B) $\lim\limits_{x\to 4} f\left(\dfrac{x}{2}\right) = 5$

C) $\lim\limits_{x\to 2} f(x^2) = 10$

D) $\lim\limits_{x\to 2} (f(x))^2 = 10$

10. The figure shows the graph of function f. Which of the following statements best describes f?



A) Since $\lim\limits_{x\to -2^-} f(x) \neq \lim\limits_{x\to -2^+} f(x)$, f has an infinite discontinuity at $x=-2$.

B) f is discontinuous at $x=0$ because $\lim\limits_{x\to 0^-} f(x)$ does not exist.

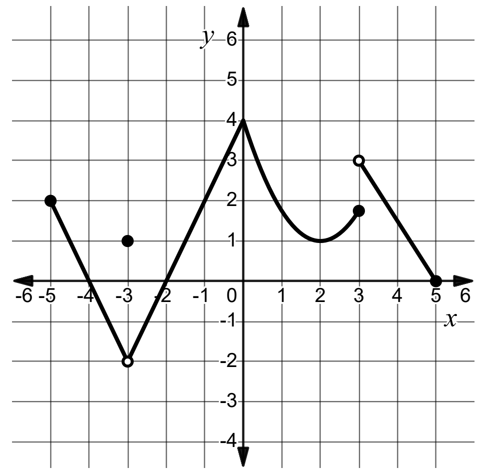
C) Function f has a removable discontinuity at $x=1$ because $\lim\limits_{x\to 1} f(x) \neq f(1)$.

D) Since $\lim\limits_{x\to 4} f(x)$ does not exist, f is discontinuous at $x=4$.

AP Calc Lesson 1.8 Homework

Name ______________________

1. The graph of $y = f(x)$ is shown. Abigail, Benny, and Chanda are discussing the function's behavior at $x = -3$.



- Abigail says that f is continuous at $x = -3$ because $f(-3)$ is defined.
- Benny says that if $f(-3)$ was assigned the value of -2, f would be continuous.
- Chanda says that f is not continuous at $x = -3$ and the discontinuity can't be removed because it is a jump discontinuity.

Which of these students, if any, is correct? Explain.

2. Let $h(x) = \begin{cases} \dfrac{x^2 + 6x - 27}{x + 9} & x \neq -9 \\ k & x = -9 \end{cases}$ for some constant k. If h is continuous for all real numbers, find the value of k.

3. Consider the piecewise function h defined below. Determine all x-value(s) where h is discontinuous and classify each as removable or non-removable.

$$h(x) = \begin{cases} 2x - 1 & x < -2 \\ x^2 - 9 & -2 \leq x < 1 \\ 3x + 2 & 1 \leq x < 3 \\ 4x - 1 & x > 3 \end{cases}$$

4. Let q be the piecewise function given by

$$q(x) = \begin{cases} x - 4 & x \leq 1 \\ 2x^2 + ax - 8 & x > 1 \end{cases}$$

Find the value for a that makes q continuous at $x = 1$.

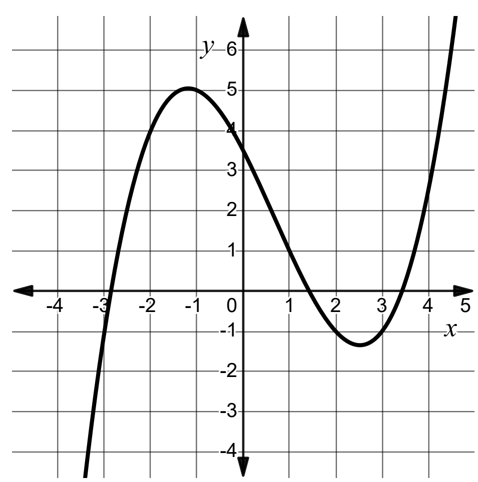
5. Let f be a function with exactly one discontinuity at $x = 2$. Selected values of f are given in the table.

x	1.8	1.9	1.99	1.999	2	2.001	2.01	2.1	2.5
$f(x)$	3.453	3.718	3.971	3.997	-4	4.003	4.029	4.298	5.640

Is the discontinuity at $x = 2$ removable? Justify your answer.

6. Let f be a continuous function defined by $f(x) = \begin{cases} a\cos\left(\frac{\pi}{6}x\right) + 1 & x \leq 4 \\ 3^{\sqrt{x}} - \frac{1}{2}x & x > 4 \end{cases}$. Find the value of a .

7. A table of values is given for a continuous function h . The graph of $y = f(x)$ is shown. Let g be the function given by $g(x) = \begin{cases} h(x) & x \leq 2 \\ f(x) + k & x > 2 \end{cases}$. Find the value of k so that g is continuous.



x	$h(x)$
-1	3.25
2	-4
3	-5
5	1

8. Let f be the function defined below, for some constant b. Find a value b so that f is continuous for $x > 0$ or explain why no such value exists.

$$f(x) = \begin{cases} \dfrac{e^{3x}}{2\sqrt{5x}} & 0 < x < 1 \\ 3 - b\cos(x^2 - 2x + 3) & x \geq 1 \end{cases}$$

9. Let g be a continuous function where

$$g(x) = \begin{cases} \dfrac{x^2 - 10x + 24}{a(x-4)} & x \neq 4 \\ a & x = 4 \end{cases}$$

Find the value of a.

A) -2

B) $\sqrt{2}$

C) 6

D) No such value exists.

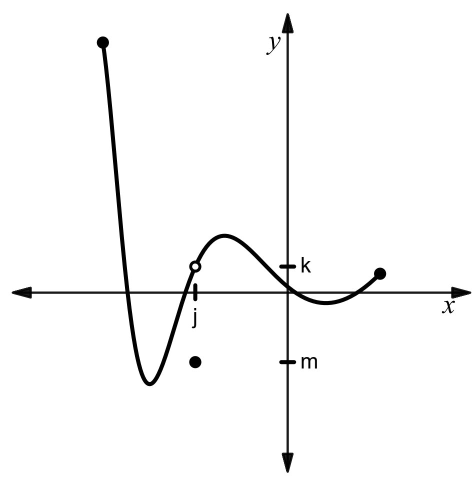
10. The complete graph of $y = f(x)$ is shown for $-2 \leq x \leq 1$. Which of the following functions is continuous on the open interval $(-2, 1)$?

A) $g(x) = \begin{cases} f(x) & x \neq k \\ j & x = k \end{cases}$

B) $g(x) = \begin{cases} f(x) & x \neq j \\ m & x = j \end{cases}$

C) $g(x) = \begin{cases} f(x) & x \neq j \\ k & x = j \end{cases}$

D) $g(x) = \begin{cases} f(x) & x = k \\ m & x \neq k \end{cases}$



AP Calc Lesson 1.9 Homework

Name ______________________

1. Let f be a continuous function on the closed interval $[-2, 5]$ with $f(-2) = 8$ and $f(5) = 1$. Must there be an x-value, $-2 < x < 5$, such that $f(x) = 4$? Justify your answer.

2. In 2020, the number of annual visitors to Glacier National Park was 1.7 million. In 2024, the number of annual visitors was 3.2 million. Must there have been a year between 2020 and 2024 where the number of annual visitors was exactly 2 million? Explain.

3. The temperature, in degrees Celsius, inside a greenhouse at time t, measured in hours after 6:00 AM can be modeled by a function G. At 8:00 AM, the temperature is 18˚C, and at 11:00 AM, the temperature is 30˚C.

   a. Is it reasonable to assume that G is continuous? Why or why not?

   b. Explain why there must be a time between 8:00 AM and 11:00 AM when the temperature inside the greenhouse was exactly 25˚C.

4. Let $f(x) = x^3 - 5x + 2$. Use the Intermediate Value Theorem to show that the graph of the function f must have an x-intercept on the interval $[0, 1]$.

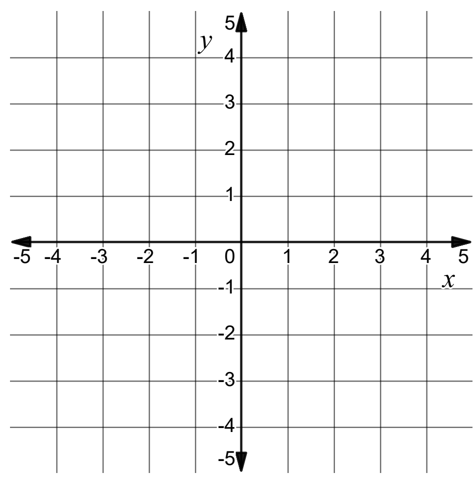
5. A rental car company charges \$0.12 per mile driven plus a \$35 booking fee. The continuous function d models the distance a particular rental car can travel, $d(g)$, on g gallons of gas. Values of $d(g)$ for selected values of g are given in the table.

g(gallons)	0	2	3	4	5
$d(g)$(miles)	0	48	63	80	92

a. Write an equation for the total cost of this rental car, C , when driving $d(g)$ miles.

b. Determine whether there must exist a value of g in the interval $2 \leq g \leq 5$, for which the total cost of the rental is \$42. Justify your answer.

6. Let g be a function with $g(-4) = -3$ and $g(4) = 2$. It is not guaranteed that there exists a c for $-4 < c < 4$ such that $g(c) = 1$. Sketch a possible graph of g .



7. The function g is defined as $g(x) = \begin{cases} x^2 - 10 & x \le 4 \\ \sqrt{x+5} & x > 4 \end{cases}$

a. Is g continuous on the interval $[0, 8]$?

b. Can the Intermediate Value Theorem be used to show that $g(x) = 0$ has a solution on the interval $[0, 8]$. Explain why or why not.

8. The rate at which water flows into a tank, in liters per minute, is given by $R(t) = 15e^{-0.1t} + 2\sin\left(\frac{\pi t}{6}\right)$. The rate at which water flows out of the tank, in liters per minute, is given by $W(t) = 3 + \ln(t+1)$.

a. Are we guaranteed a time t for $0 \le t \le 10$ where the rate at which water flows into the tank is exactly equal to the rate at which water flows out of the tank? Justify your answer.

b. If it is possible, find the time t at which it occurs. If it is not possible, state that it is not possible.

9. Selected values of a continuous function h are given in the table. For $0 \leq x \leq 11$, what is the minimum number of solutions to the equation $h(x) = 2$?

x	0	2	4	7	11
$h(x)$	-2	1	3.5	5.5	1.25

A) One

B) Two

C) Three

D) Four

10. Let f be a function such that $f(-3) = 10$ and $f(5) = 4$. Which of the following statements could be used to justify that there must be a value x where $-3 < x < 5$ such that $f(x) = 6$.

A) f is defined on the entire interval $(-3, 5)$

B) f is continuous on the entire interval $[-3, 5]$

C) f is decreasing on $[-3, 5]$

D) $\lim\limits_{x \to c^-} f(x) = \lim\limits_{x \to c^+} f(x)$ for all c on $(-3, 5)$